Project details

Log-Log Legends · DTU 02805 · state on 14 November 2026 · the one-page version is *overview.pdf*. Each analysis gets a short block: what it asks, where it lives, what it found.

Data and shared rules

What	Count	Source
Movies with 10,000+ votes (the graph)	12,673	IMDb
Clean films with budget and gross in USD	2,446 (100k+), 6,851 (10k+)	Wikipedia infoboxes
Screenplays	1,095 clean 100k+ films	MovieSum
Audience reviews	2,070 clean 100k+ films, to 2017	MovieLens Tag Genome
Critic scores	Metacritic 2,431, RT 2,740	Wikipedia, Wikidata

"Clean" means budget and gross both in USD and at least $1,000, no rentals-only or regional gross, no streaming distributor, and released before 2026. The 100k+ tier favours hits: only 169 of its 2,446 films grossed less than their budget. Profit is ln(gross/budget) everywhere. Network measures use only earlier films, every result gets a null or a second method, and no script text, review text or IMDb fields go into git.

Who made the film

Movie graph (`scripts/network/`, `notebooks/movie_graph.ipynb`)

Movies, people (by role and billing) and companies in one network, with studio names merged through Wikidata IDs.

- The studio's track record is the strongest single measure: $\rho = +0.27$ with the multiple, net of budget and year, on 4,568 films.
- All network measures together lift R^2 from 0.041 to 0.130 over budget, year and genre (shuffle $p = 0.005$).
- A director's next film is a hit 35% of the time after a hit, and 18% after a non-hit.

Shared-people returns (`scripts/network/roi_similarity.py`)

Do films with the same director, producer, cast member or studio have more similar returns than chance?

- Every role clusters: 0.09 (top-three cast) to 0.17 (producers), against a shuffle null of at most 0.02.
- It survives dropping franchise films, ranking the returns, and keeping only films from 1980 on. About half survives a shuffle within each studio.
- It's descriptive: it uses all years at once.

Actor network (Niklas, branch `initial_actor_network`, not merged)

Top-three billed actors, linked by shared films. It comes with an interactive page with projections, communities, centrality and per-actor ROI.

What the film says

Scripts and reviews notebook (`notebooks/scripts_and_reviews.ipynb`)

- Only 2 of 15 script features pass $q < 0.05$ net of budget and year. The strongest is a character network centred on one character ($\rho = -0.097$).
- Six VADER arc shapes don't differ in profit ($p = 0.99$).

Scattertext words (`analysis/scripts_text/`, pre-registered)

Top against bottom quarter by multiple, within genre and decade, on dialogue.

- The difference is slightly above random groupings ($z = 2.08$, $p = 0.024$).
- It disappears without franchise films ($z = 0.50$). Dropping the same number of random films never does that (0 of 50 tries), so the signal sits in the franchise films themselves.
- A word classifier doesn't beat genre, decade and budget (AUC 0.58 against 0.58).

Emotional arcs (`analysis/arcs/`, pre-registered, one test)

Every sentence of 834 non-franchise scripts is scored by DistilBERT SST-2. Each film becomes a 100-point tone curve, summarised as swing (D1) and ending lift (D2). Budget, genre, decade and scene length are controlled.

- Swing: no link ($\rho = +0.035$, $p = 0.32$).
- Ending lift: a near miss ($\rho = +0.078$, $p = 0.028$, against a pre-set cutoff of 0.025). Its sign holds in 12 of 13 variants, but not with VADER.

- The arc adds nothing to out-of-sample prediction.
- About half of the arc variation between films is scene noise (split-half reliability around 0.45).
- A fourth sentiment model, ModernBERT, is running as a robustness check.

How it was received

Reviews exploration (Àngela, branch `reviews-exploration`, not merged)

2,095 films up to 2017. Every correlation is tested against a shuffle within genre × decade.

- The multiple goes with IMDb rating at $\rho = 0.27$, Metacritic 0.26 and Rotten Tomatoes 0.33. Genre and decade alone give about 0.1.
- The critics agree with each other far more (Metacritic and Rotten Tomatoes at 0.93). Review tone goes with the multiple at 0.13.
- A disagreement score (rating rank minus money rank) finds "liked beyond their money" films (*tarantino*, *noir*) and "earned beyond their liking" films (*scares*, *slasher*).

What predicts profit (`notebooks/what_predicts_profit.ipynb`)

Trained on films to 2010, tested on 2011 to 2017.

Model	Out-of-sample R^2
Budget, year, genre	−0.010
+ network	0.202
All pre-release blocks (adds month, franchise)	0.244
+ critic and audience scores	0.312

- On films with a screenplay, script features add nothing beyond the network (R^2 0.037 against 0.036).
- Director hit streaks beat a shuffle of each director's own films (+0.175 against +0.090). PageRank adds nothing beyond plain experience.
- Rotten Tomatoes goes with the multiple at $\rho = +0.25$. There are 158 loved flops and 205 panned hits.

Open and weak spots

- **Human validation**, which only people can fill:
  - Scattertext: `analysis/scripts_text/outputs/validation_*.csv`
  - Arcs: `analysis/arcs/validation/` and `analysis/arcs/outputs/s12_back_to_text.csv`
  - Reviews: the 50-review sample on `reviews-exploration`
- **Arcs Step 11** waits on the instrument question (whole-review VADER against sentence-level VADER).
- **Weak spots**: the franchise flag is rough, release month isn't checked, reviews stop at 2017, and gross over budget isn't profit.

Where things are

`scripts/`	Data download and parsing, the graph, network features
`analysis/scripts_text/`, `analysis/arcs/`	The two pre-registered text analyses: config, LOG, steps, outputs
`notebooks/`	Graph, profit model, scripts and reviews, one folder per text analysis
`site/scripts/`, `site/arcs/`	Static pages for the text analyses
`data/bundles/`	Compressed data that ships with the repo